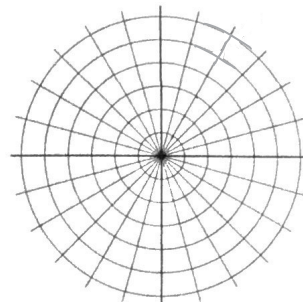


1. a) On the axes below, sketch point P that has polar coordinates: $(5, \frac{-5\pi}{3})$ [1pt]

b) Rename point P with a positive angle and negative radius _____ [1pt]

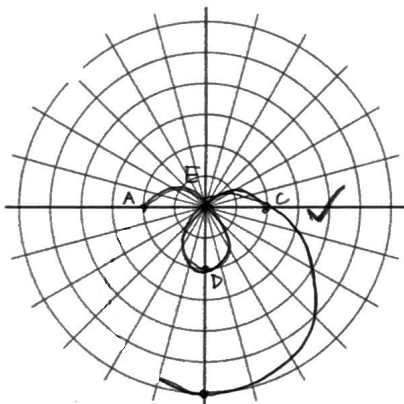
c) Convert point P into rectangular coordinates (simplified) _____ [1pt]



2. Convert the rectangular point $(0, -100)$ into simplified polar coordinates

_____ [2pts]

3. Accurately graph polar curve $r = -2 - 4\sin\theta$ by making a table of values. **Label** at least four **different** points on your graph using letters A, B, C, and D and show their polar coordinates in the table below. You may use the additional columns in the table to determine more points (this is optional) [6pts]



Name of point	A	B	C	D		
θ						
r						

4. The graph of $r = 2 + 6\cos 3\theta$ has an interesting shape. The diagram below (not drawn to scale) shows a part of the curve, including a “petal inside a petal” along the angle $\theta = 0$.

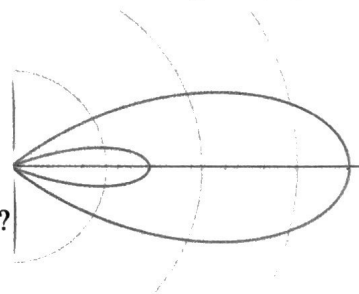
a) How long is the longer petal along $\theta = 0$? _____ [1pt]

b) How long is the shorter petal along $\theta = 0$? _____ [1pt]

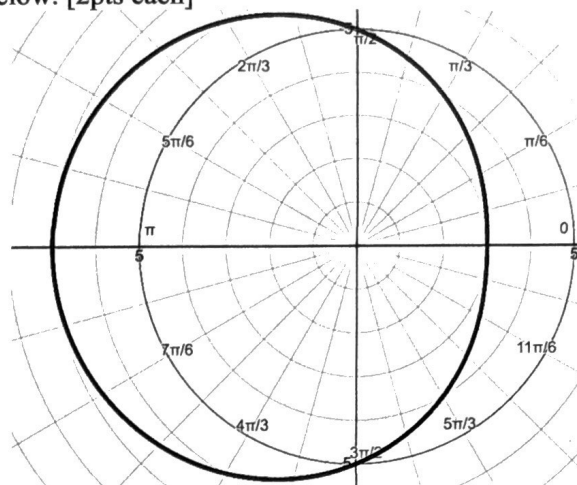
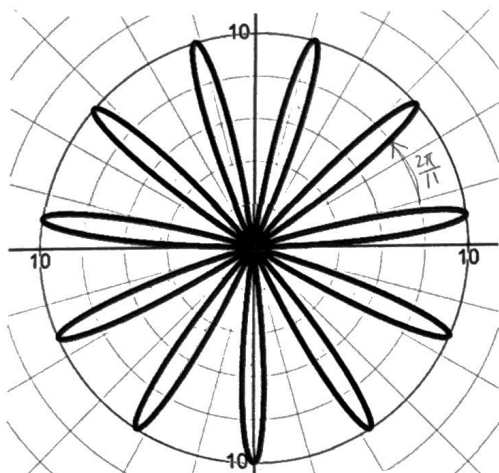
c) Along what other angles θ (besides $\theta = 0$) will “petal in petal” also occur?

List all such angles θ in radians. _____ [2pts]

d) For what angles $0 \leq \theta \leq \frac{\pi}{2}$ will this curve intersect the circle $r = 2$? Show all the work that leads to your answer(s). [3pts]



5. a) Write polar equations that match the two graphs below. [2pts each]



Answer the following questions about the polar rose graphed above on the left.

b) How many radians will it take to create a single petal? _____ [1pt]

c) Give a polar equation of the first line of symmetry in the first quadrant. _____ [1pt]

6. Derive a general polar equation of a **horizontal** line that passes through the polar point (k, α) (where both k and α are constants). Give your answer in terms of k , α and θ . [2pts]

$r =$ _____

7. Convert the equation: $r = \frac{3}{1-\sin\theta}$ into rectangular form. Rewrite your answer (without a square root sign) to simplify it into a form of a common function. What is the shape of the graph? [4pts]

Rectangular form: _____. What shape is the graph? _____